

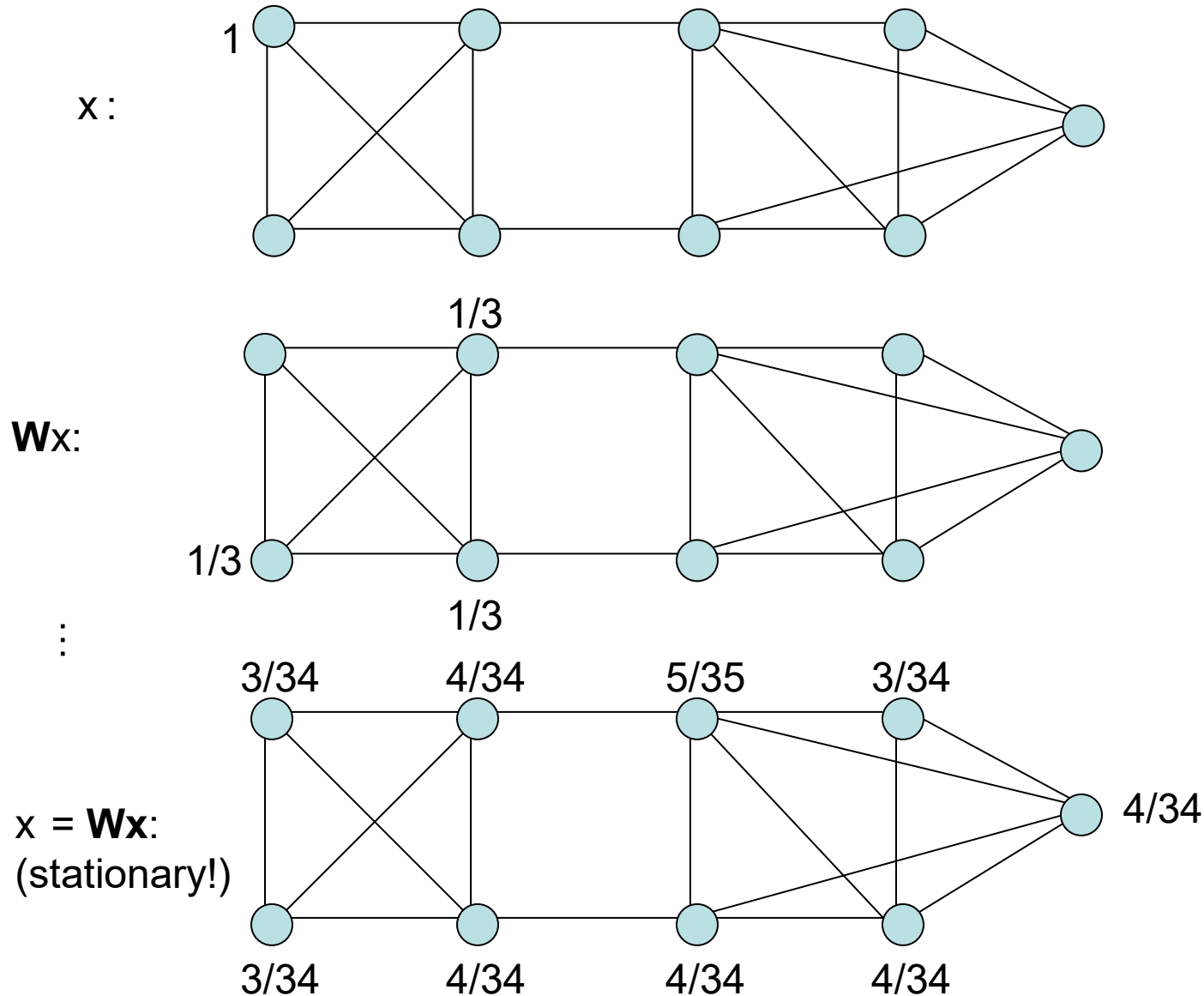
Eigenvectors and Stationarity

- Given an $n \times n$ matrix $\mathbf{A}$, we say $(\lambda, \mathbf{x})$ is an (eigenvalue, eigenvector) pair of $\mathbf{A}$ if $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$.
(So $\mathbf{A}$ scales $\mathbf{x}$, but does not change its direction)
- An $n \times n$ matrix generally has n eigenvalue-eigenvector pairs.
- Let's think of an eigenvector $\mathbf{x}$ as a “stationary” distribution with respect to multiplication by $\mathbf{A}$...
- If $\mathbf{A}$ is symmetric, these are particularly well-behaved: real-valued and orthogonal to each-other.
- Rectangular matrices don't have eigenvectors, but instead have left and right “singular vectors”: eigenvectors of the symmetric matrices $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}\mathbf{A}^T$ (these have the same eigenvalues).

The Power Method

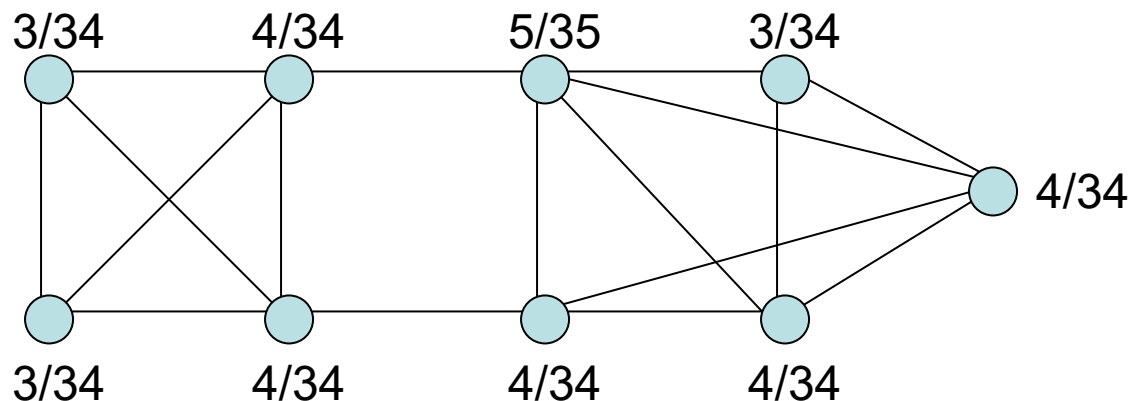
- Nice interpretation of the action of A on a vector x to produce Ax as stretching components of x along directions of eigenvalues by scale factors given by eigenvalues.
- Easy to calculate *principal* eigenvector (with largest eigenvalue):
Start with an x and repeat {
 $x = Ax$
 Normalize x to have unit length: $x = x / ||x||$
}
- Conveniently many eigenvector-based methods in data analytics require only the principal eigenvector.

Random Walks



W is the “walk” matrix of our graph (W_{ij} = probability of stepping to i if at node j)
Principal eigenvector x of W has eigenvalue 1, so $Wx = x$ (stationary distribution)
(* as long as graph is connected and not bipartite)

Random Walks : Applications



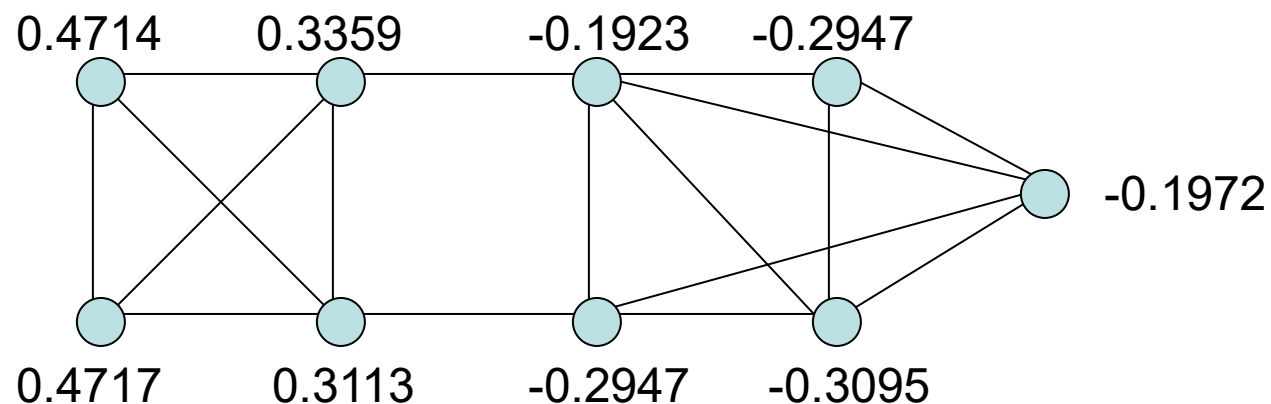
Principal eigenvector of **W**: nonnegative weights reflecting **centralities**

Undirected graph: just node degrees

Directed graph: Google Pagerank scores! (with addition of teleportation)

Other eigenvectors: Positive and negative weights reflect **cluster pairs**

Solution of relaxation of NP-hard “Normalized Cut” problem, asking for pair of clusters (A,B) minimizing $\text{Prob}[\text{random walk crosses } A \leftrightarrow B]$.

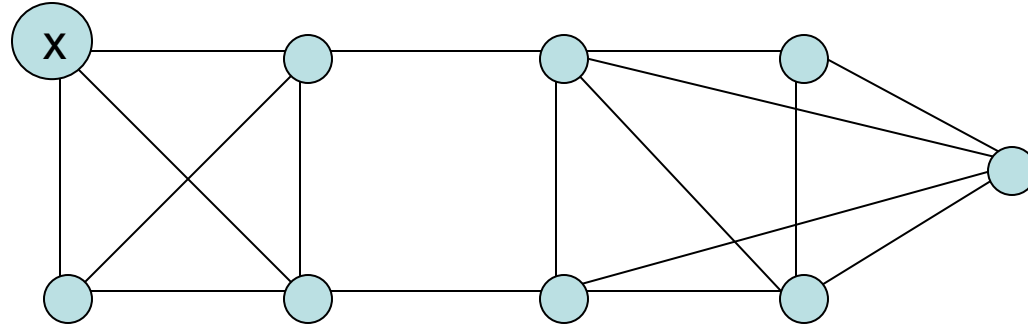
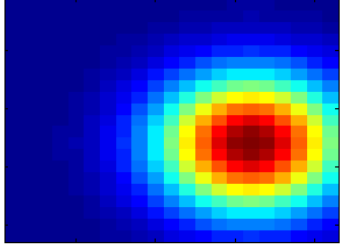


Further Extensions

- HITS algorithm: similar analysis but separates outbound (hub) and inbound (authority) behavior of each node in a directed citation network.
 - Principal eigenvector: nonnegative hub and authority scores
 - Positive and negative ends of other eigenvectors: cluster pairs of hubs and authorities that mutually reinforce each-other.
- Image segmentation:

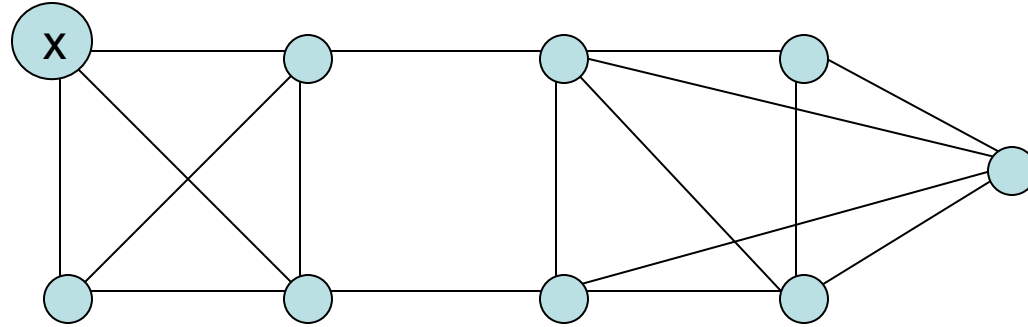
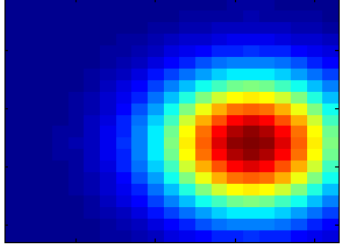


Diffusion Kernels / Personalized Pagerank



- How can we find a distribution of weight in a graph that feels like a Gaussian centered on node x ?

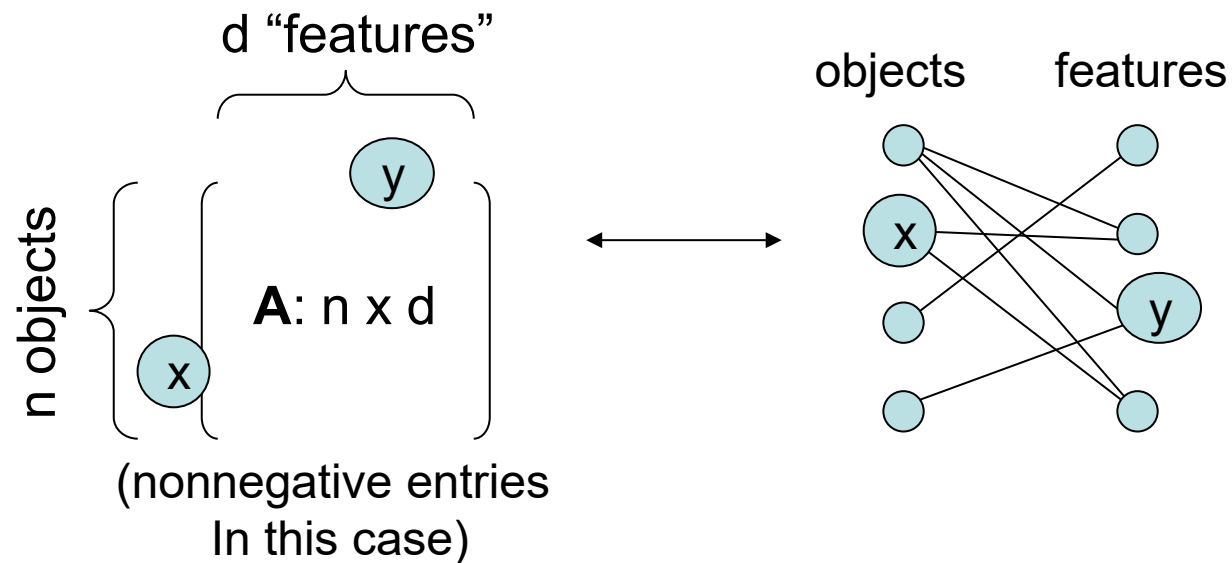
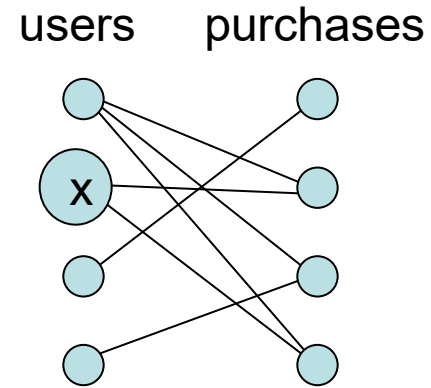
Diffusion Kernels / Personalized Pagerank



- How can we find a distribution of weight in a graph that feels like a Gaussian centered on node x ?
- Do a random walk but each step has $\sim 10\%$ probability of teleporting back to x .

(compare to normal Pagerank, where each step has $\sim 10\%$ probability of teleporting to random location)

Applications: Recommendation / Imputation of Missing Data



Quadratic Form Minimization

- Consider maximizing or minimizing a “quadratic form”:
 $x^T \mathbf{A} x$ (with $\mathbf{A}$ square and symmetric) or
 $x^T \mathbf{A} y = \sum_{ij} \mathbf{A}_{ij} x_i y_j$ (with $\mathbf{A}$ rectangular),
where x and y are constrained to have unit length.
(think of $x^T \mathbf{A} x$ as just the matrix version of “ ax^2 ”)
- Optimal solution is to pick x (and y) as highest or lowest eigenvectors (singular vectors, in the rectangular case) of $\mathbf{A}$.
- Many problems (e.g., PCA, modularity cut, ratio cut, normalized cut, spectral embedding) can be cast in this form or relaxed to this form, and so have eigenvectors as solutions.

Example: 1D Embedding

- Assign a location x_i on the number line to each node / data point i such that we minimize:

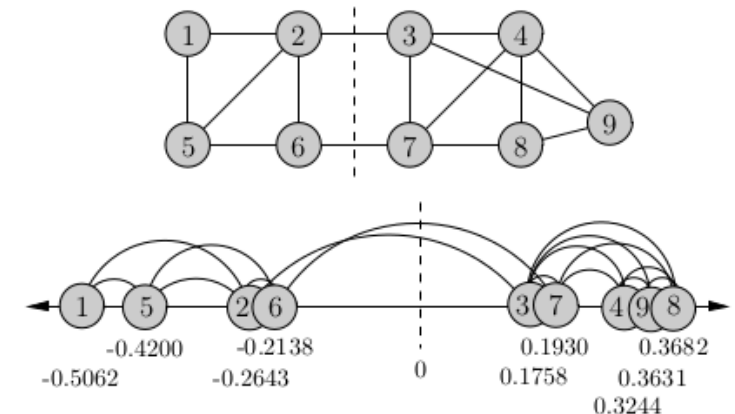
$$f(x) = \sum_{i,j} \mathbf{A}_{ij} |x_i - x_j|^2,$$

where x has zero mean and unit variance.

- Can re-write $f(x)$ as quadratic form minimization: $f(x) = 2x^T \mathbf{L}x$, where $\mathbf{L}$ is the **Laplacian** matrix of $\mathbf{A}$

Diagonal entries: $\mathbf{L}_{ii} = \sum_j \mathbf{A}_{ij}$ Non-diagonal entries: $\mathbf{L}_{ij} = -\mathbf{A}_{ij}$

- $\mathbf{L}$'s lowest eigenvalue is 0 with eigenvector all 1s. Can't choose it, since we want zero mean. Solution: 2nd lowest eigenvector, which can be computed with a modified power method.



Two-Dimensional Example

